

Differential Calculus — Logarithmic Differentiation

Chapter 2: Exponential-Trigonometric Product Function | Page 34

PROBLEM STATEMENT

Logarithmic Differentiation & Chain Rule

Find $\frac{dy}{dx}$ from the given function:

$$y = e^{\sin x} \sin(a^x)$$

STEP 1 প্রদত্ত যৌগিক গুণফল ফাংশনটি উল্লেখ করি (Given Function)

$$y = e^{\sin x} \sin(a^x)$$

STEP 2 উভয়পক্ষে স্বাভাবিক লগারিদম (log) গ্রহণ করে গুণফলকে যোগফলে রূপান্তর করি (Taking Natural Logarithm on Both Sides)

$$\log y = \log [e^{\sin x} \cdot \sin(a^x)] = \log(e^{\sin x}) + \log(\sin(a^x)) = \sin x + \log(\sin(a^x))$$

STEP 3 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Differentiating w.r.t. x)

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx}(\sin x) + \frac{d}{dx}[\log(\sin(a^x))]$$

STEP 4 চেইন রুল প্রয়োগ করে প্রতিটি অংশের অন্তরজ নির্ণয় করি (Applying Chain Rule & Standard Formulas)

$$\frac{1}{y} \frac{dy}{dx} = \cos x + \frac{1}{\sin(a^x)} \cdot \cos(a^x) \cdot \frac{d}{dx}(a^x) = \cos x + \cot(a^x) \cdot (a^x \log a) = \cos x + a^x \cot(a^x) \log a$$

STEP 5 উভয়পক্ষকে y দ্বারা গুণ করে চূড়ান্ত অন্তরজ পাই (Multiplying by y for Final Derivative)

$$\frac{dy}{dx} = y [\cos x + a^x \cot(a^x) \log a] = e^{\sin x} \sin(a^x) [\cos x + a^x \cot(a^x) \log a]$$

FINAL ANSWER

$$\frac{dy}{dx} = e^{\sin x} \sin(a^x) (\cos x + a^x \cot(a^x) \log a)$$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- Logarithmic Differentiation:** $\frac{d}{dx}[\ln y] = \frac{1}{y} \frac{dy}{dx}$. Converts products into simple additions.
- Exponential Base \$a\$ Derivative:** $\frac{d}{dx}[a^x] = a^x \ln a = a^x \log a$.
- Logarithm of Exponential:** $\log(e^{\sin x}) = \sin x \log e = \sin x$.
- Cotangent Identity:** $\frac{\cos(a^x)}{\sin(a^x)} = \cot(a^x)$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Base \$a\$ Exponent Differentiation:** Misapplying the power rule $\frac{d}{dx}(a^x) \neq xa^{x-1}$. The base is constant while the exponent is variable!
- Missing Factor of \$y\$:** Solving for $\frac{1}{y} \frac{dy}{dx}$ and forgetting to multiply through by y in the final step.
- Argument Nesting:** Writing $\sin a^x \cdot \cot a^x = \cos a^x$ directly if expanded via product rule.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Base Restriction:** The exponential function a^x requires $a > 0$ and $a \neq 1$.
- Real Domain:** Under standard real analysis, $\text{Domain} = (-\infty, \infty) = \mathbb{R}$.
- Differentiability:** Infinitely smooth (C^∞) everywhere on $\mathbb{R}$.

BOX 4: ALTERNATIVE METHOD VIA DIRECT PRODUCT RULE

- Using $(u \cdot v)' = uv' + vu'$:

$$\frac{dy}{dx} = e^{\sin x} \cdot \frac{d}{dx}[\sin(a^x)] + \sin(a^x) \cdot \frac{d}{dx}[e^{\sin x}]$$

- Differentiating each term:

$$\frac{dy}{dx} = e^{\sin x} \cdot [\cos(a^x) \cdot a^x \log a] + \sin(a^x) \cdot [e^{\sin x} \cdot \cos x]$$

- Factoring out $e^{\sin x} \sin(a^x)$:

$$\frac{dy}{dx} = e^{\sin x} \sin(a^x) \left[a^x \frac{\cos(a^x)}{\sin(a^x)} \log a + \cos x \right] = e^{\sin x} \sin(a^x) (\cos x + a^x \cot(a^x) \log a)$$

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1–2 (Logarithmic Transformation):** 1.0 Mark
- Step 3 (Implicit Logarithmic Differentiation Setup):** 1.0 Mark
- Step 4 (Trig & Base-\$a\$ Exponential Derivative):** 1.0 Mark
- Step 5 (Resubstitution of \$y\$ & Final Form):** 1.0 Mark
- Total Allocation:** 4.0 Marks